

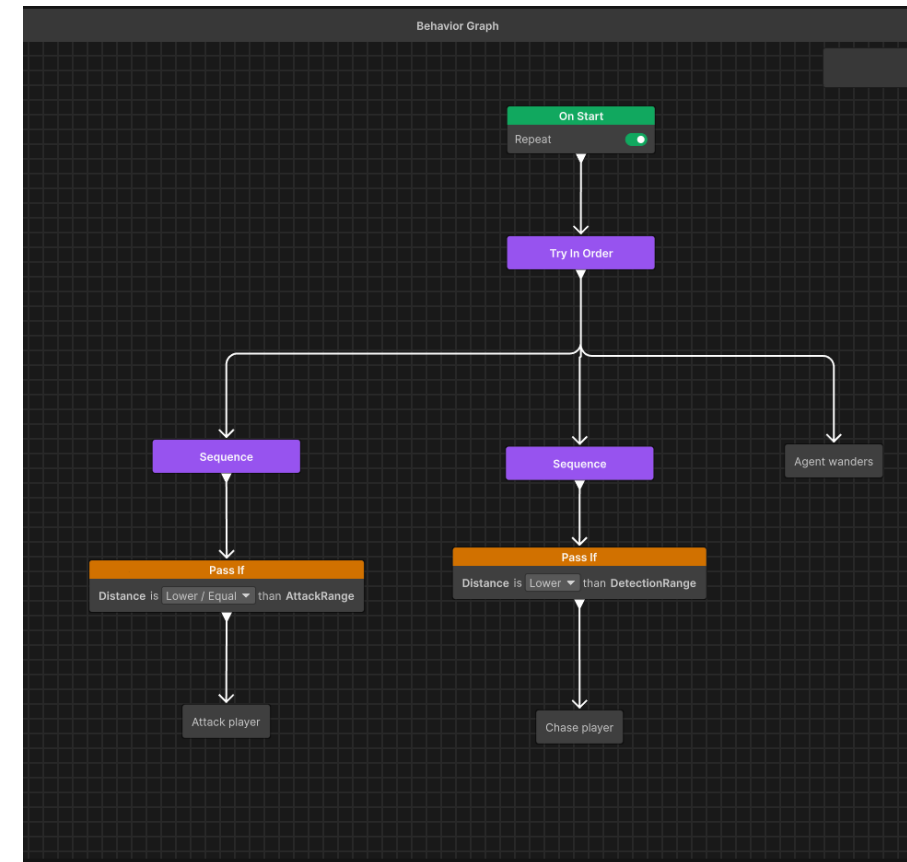
AI for games

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- 20098883
- CA3 - Integrated AI Project + Evaluation



AI System Overview

- Behaviour Tree controls decision making
- Blackboard stores distance to player
- Conditional distance checks
- A* pathfinding for movement
- FSM-style scripts execute actions





Unity AI System

- NavMesh AI navigation system
 - Behaviour graph / custom logic
 - Blackboard manually implemented
 - Flexible but low level system
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Unreal AI System

- Built in behaviour trees
 - Native blackboard system
 - AI perception
 - More structured AI pipeline
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Unity vs Unreal Differences

Unity

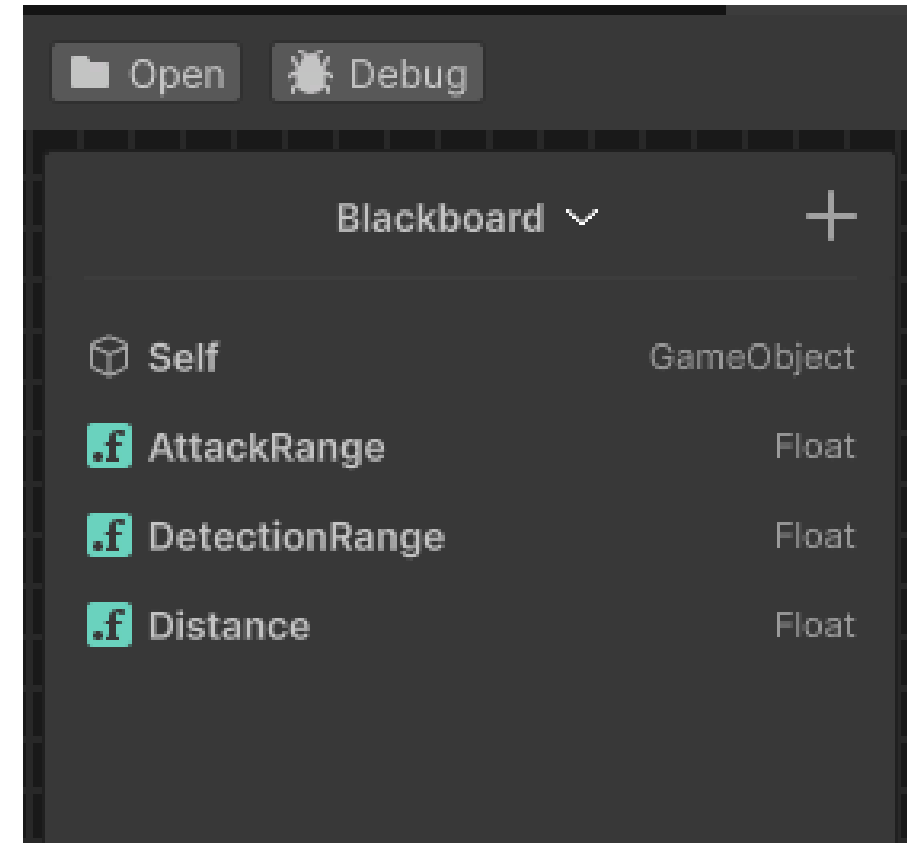
- Modular, code driven
- Manual setup
- Gives control

Unreal

- Integrated AI framework
 - Editor driven workflow
 - Gives structure
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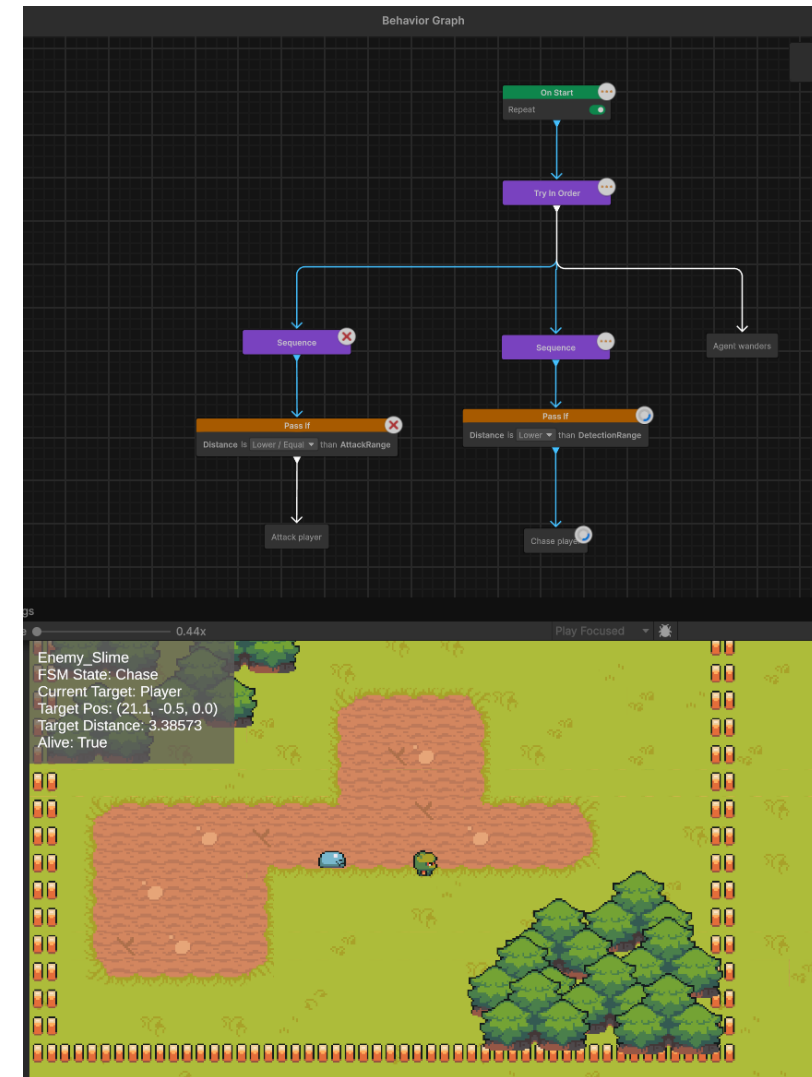
My AI Flow

- Player distance calculated each frame
- Stored in blackboard
- Behaviour tree checks thresholds
- Actions executed by scripts



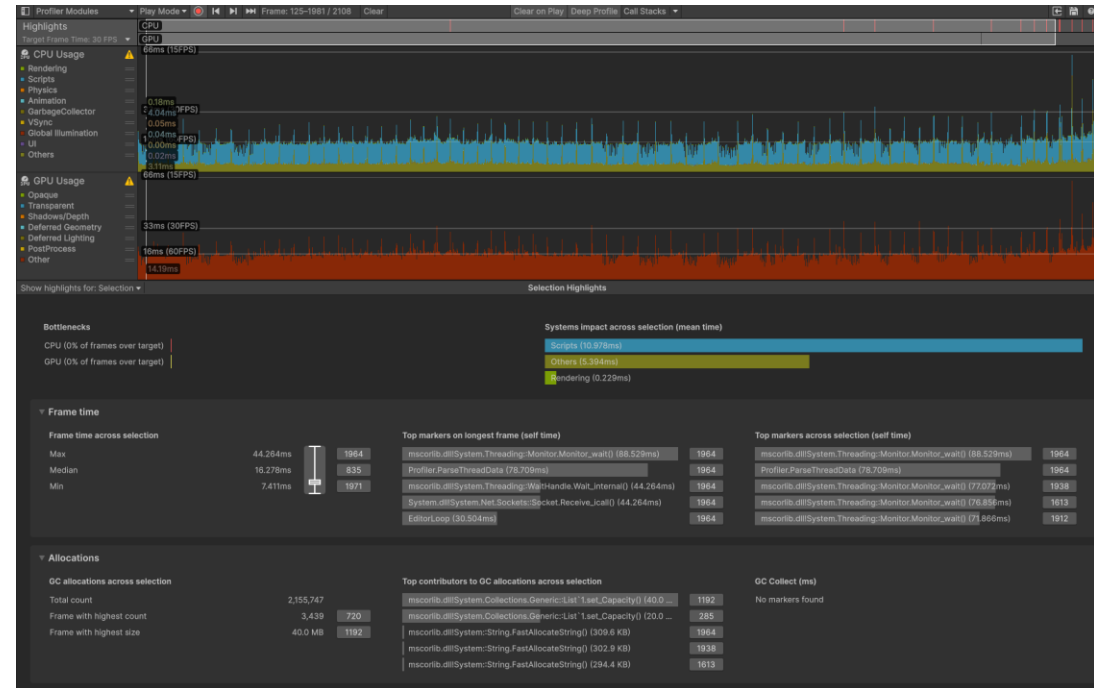
Behaviour States

- Wander: Random movement
- Chase: A Star pathfinding to player
- Attack: Timer based damage system



Performance Summary

- Chase: Highest cost (A^* recalculations)
- Attack: Small spikes (timers + checks)
- Wander: Lowest cost (infrequent updates)





Optimisation

- Pathfinding cooldown system
 - Frame spreading for A Star pathfinding
 - Distance threshold for recalculation
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Serious Games AI

- Used in training simulations (Army, pilot)
 - Behaviour must be predictable
 - AI must be consistent across runs
 - Often uses behaviour trees for clarity
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Key Constraints

- Determinism over randomness
 - Behaviour must be explainable
 - Repeatable scenarios for evaluation
 - Less focus on fun, more on reliability
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Reflection

- System interaction is critical
 - Small thresholds affect behaviour a lot
 - BT + FSM needs careful tuning
 - Future: Better perception system (alertness)
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